

Loan Default Risk Analysis

Task 7 – Critical Evaluation of Principal Component Analysis (PCA)

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1. PCA Analysis Performed

PCA was applied to 10 numerical variables from the Loan Default dataset after removing rows with missing values in the selected variables. The numerical variables were standardized before PCA so that variables measured on different scales could be considered together.

2. PCA Results

Measure	Result
Selected numerical variables	10
PC1 variance	25.79%
First 5 PCs – cumulative variance	73.88%
First 7 PCs – cumulative variance	92.17%
All 10 PCs – cumulative variance	100%

The first five principal components explain 73.88% of the total variance, while the first seven components explain 92.17%. Therefore, a smaller number of components can retain a large proportion of the variation in the selected numerical variables.

3. Scree Plot

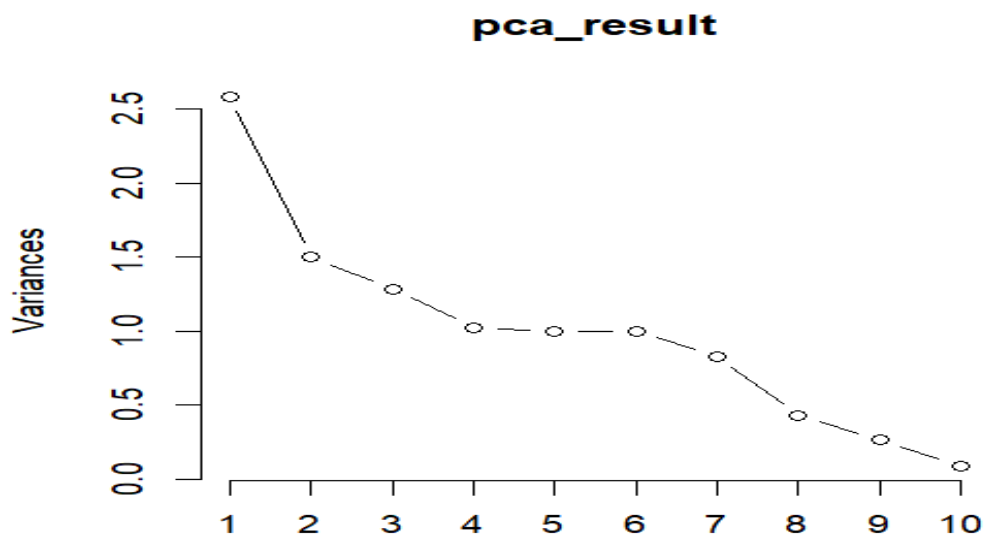


Figure 1. PCA Scree Plot

4. Is Dimensionality Reduction Necessary?

The need for dimensionality reduction is limited because only 10 numerical variables were selected. PCA can still be useful if correlated variables create redundancy or if a compact predictor set is required, but it is not automatically necessary.

5. Possible Advantages of PCA

- Reduces the number of variables using a smaller number of components.

- Can reduce repeated information caused by correlated numerical variables.
- Can reduce model complexity.
- Can retain a large proportion of variance using fewer components.

6. Limitations of PCA

- Original variables become harder to interpret.
- PCA is mainly appropriate for numerical variables.
- PCA is not automatically the best method for every dataset.
- Its usefulness depends on the dataset structure.

7. Expected Impact on Predictive Modelling

PCA may reduce predictor redundancy and simplify some predictive models. It may help with multicollinearity, but PCA components are less interpretable than the original variables. Any predictive benefit should be verified using model evaluation results rather than assumed from explained variance alone.

8. When PCA Should and Should Not Be Used

PCA may be considered when there are many correlated numerical variables, when predictor reduction is important, when redundancy affects modelling, or when compact representation is preferred over direct interpretation.

PCA may be avoided when the number of numerical variables is already small, when original-variable interpretability is important, when important predictors are categorical, or when predictive models already perform well without dimensionality reduction.

9. Research Findings

The reviewed research indicates that PCA can be useful for reducing high-dimensional and correlated financial variables. One reviewed bank-lending study reduced 159 indicators to 9 principal components while retaining 84.3% of total variance. Other reviewed work indicates that PCA should not automatically be treated as the best dimension-reduction method because suitability depends on variable types, dataset structure and modelling objectives.

10. Final Critical Evaluation

The first five components explain 73.88% of the total variance and the first seven explain 92.17%. This demonstrates that PCA can reduce the number of numerical predictors while retaining most of their overall variation. However, because only 10 numerical variables were selected, PCA is not necessarily required. It should be treated as an optional modelling technique, with the final decision based on correlation structure, model performance and interpretability.